
MODEL SELECTION AND VALIDATION REPORT (pielou_biomass model)

OBSERVATIONS:3382

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (pielou_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	7.05	0.23
kde_fric	0.24	0.02
kde_feve	-0.18	0.02
tric	0.08	0.02
p_teve	-0.56	0.02
sbt_mean	0.32	0.05
sic_mean	0.05	0.05
sbt_sd	0.10	0.03
sst_sd	0.02	0.03
log_chla_mean	0.00	0.02
depth	0.13	0.03

Dispersion parameter: 0.70

Spatiotemporal AR1 correlation (rho): 0.92

Matérn range: 348.98

Spatiotemporal marginal AR1 SD: 0.80

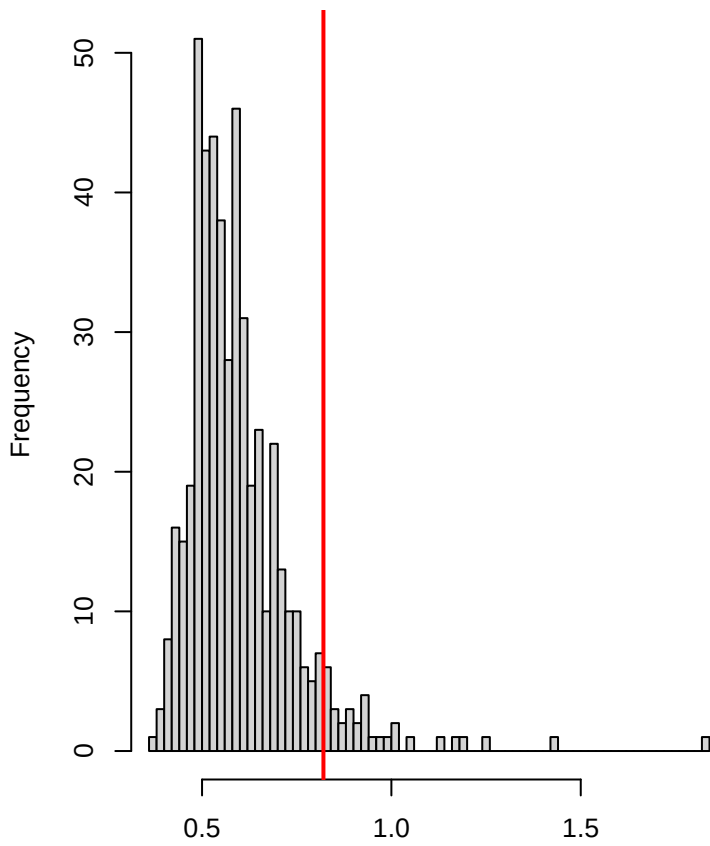
REML criterion at convergence: 26773.565

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

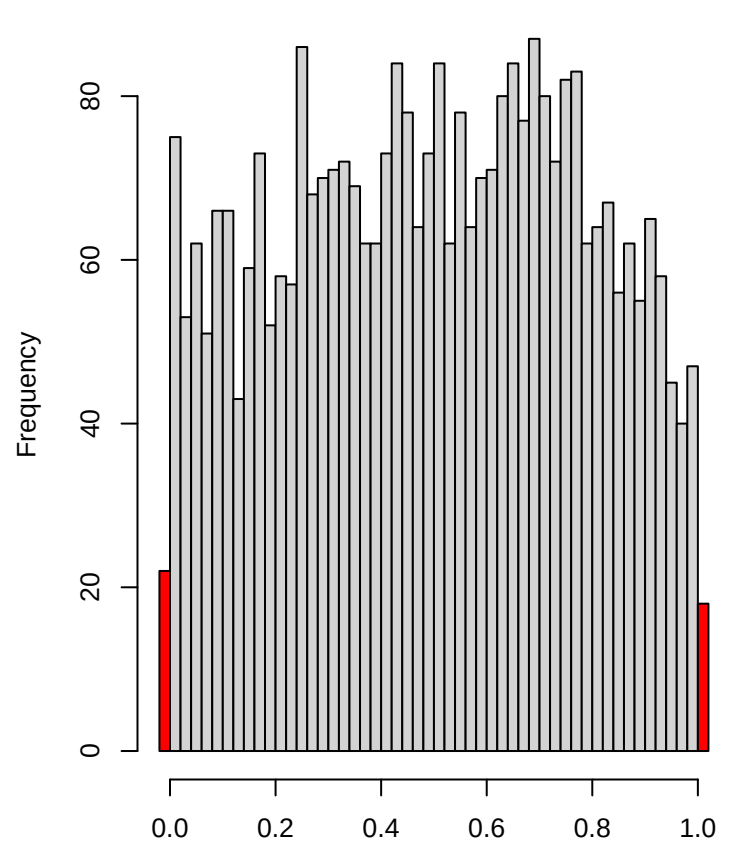
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1_with_anisotropy	17	53576.37	0.7619882
spatiotemporal_random_fields_ar1	15	53577.13	271.5360504
spatiotemporal_random_fields_iid_with_anisotropy	16	53848.67	7.9426689
spatiotemporal_random_fields_iid	14	53856.61	187.0407243
spatial_random_fields	14	54043.65	2.4209961
spatial_random_fields_with_anisotropy	16	54046.07	634.2671001
error_only	12	54680.34	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



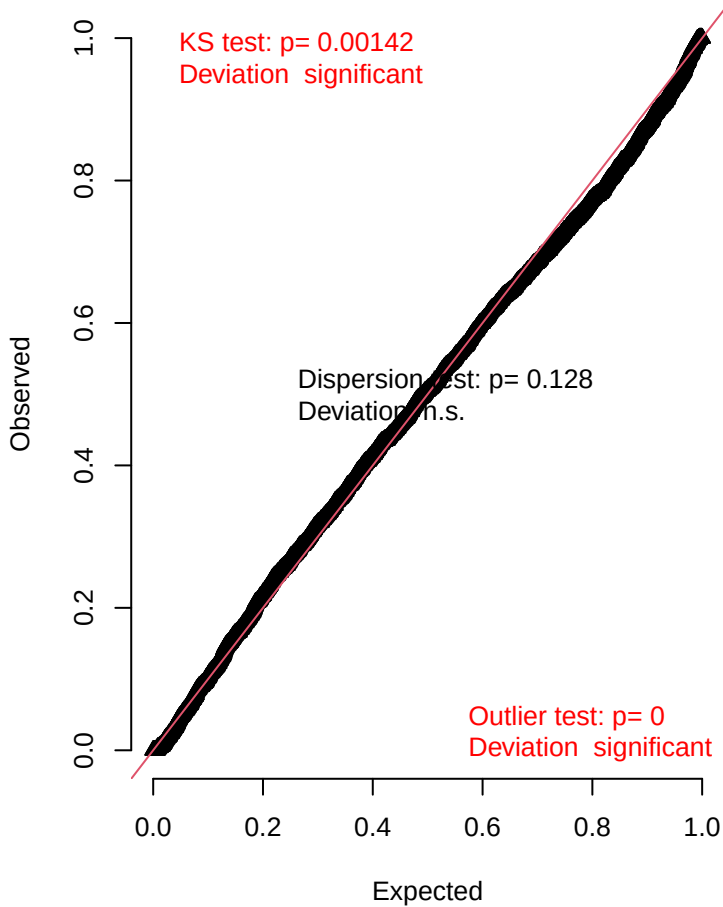
Simulated values, red line = fitted model. p-value (two.sided) =

Outlier test significant

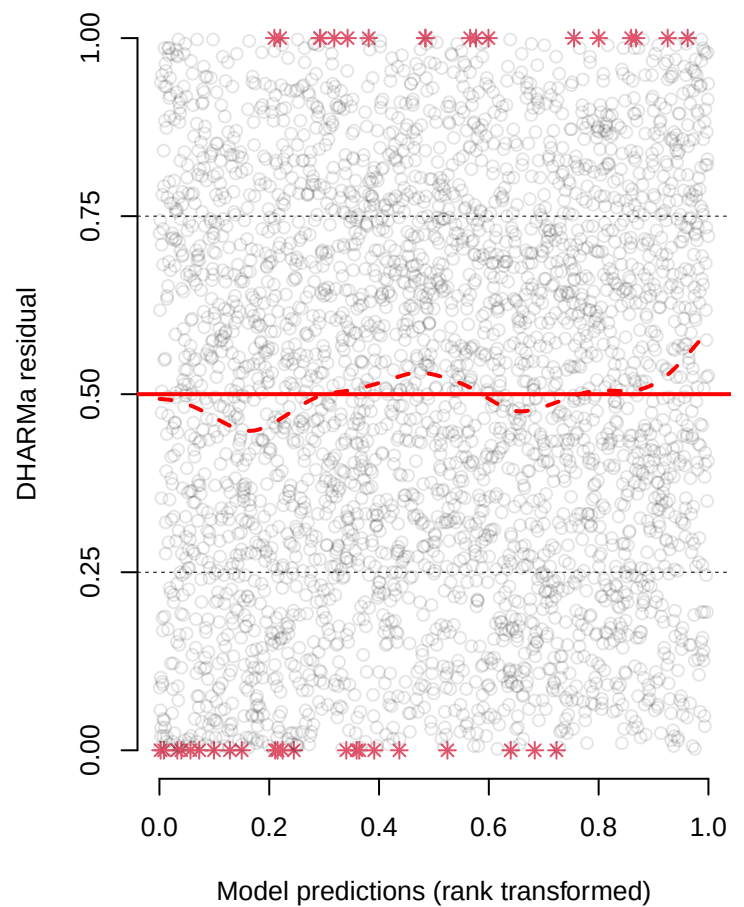


Residuals (outliers are marked red)

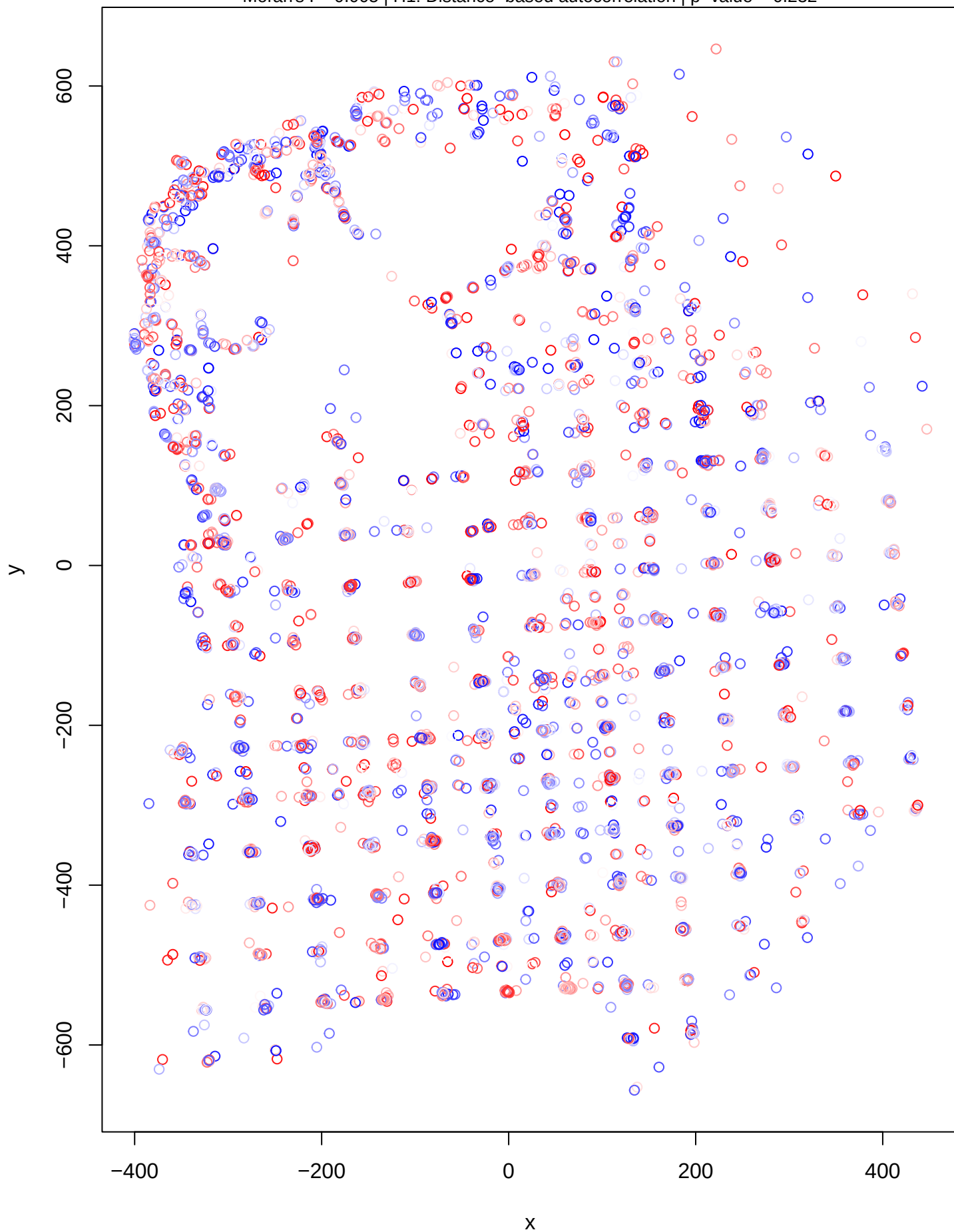
QQ plot residuals



DHARMa residual vs. predicted

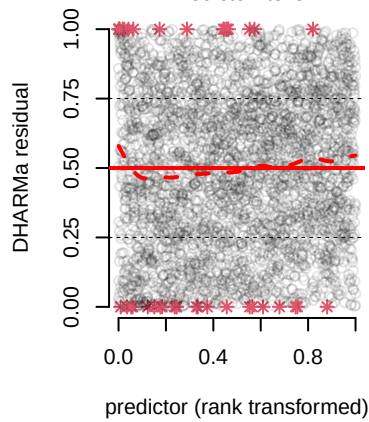


DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.005 | H1: Distance-based autocorrelation | p-value = 0.232



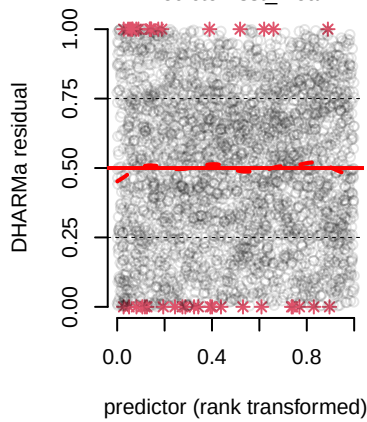
DHARMa residual Residual vs. pred

Predictor: teve



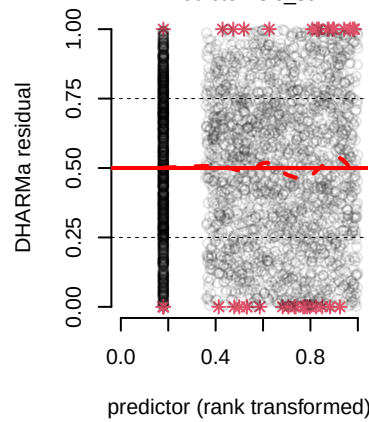
DHARMa residual Residual vs. pred

Predictor: sst_mean



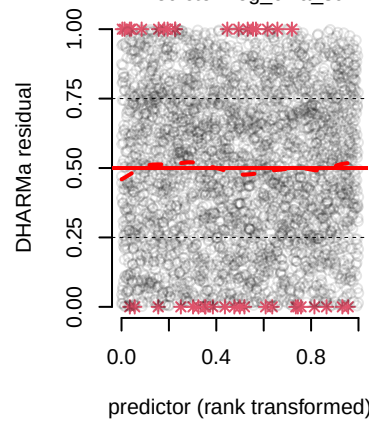
DHARMa residual Residual vs. pred

Predictor: sic_sd



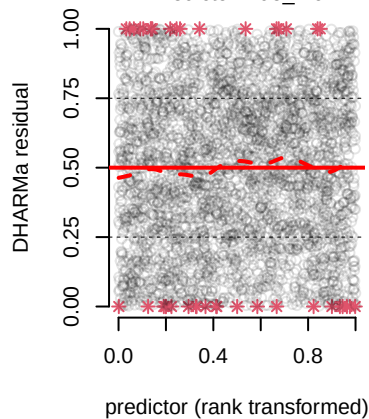
DHARMa residual Residual vs. pred

Predictor: log_chla_sd



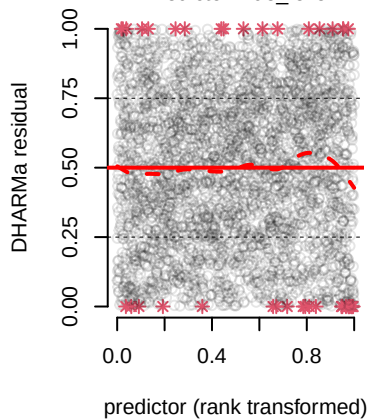
DHARMa residual Residual vs. pred

Predictor: kde_fric



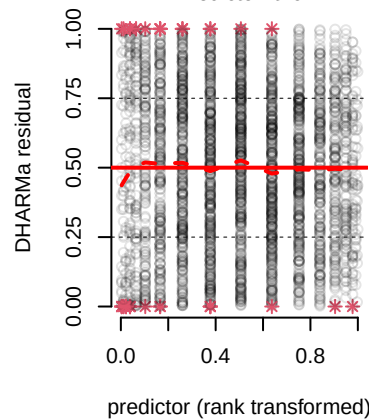
DHARMa residual Residual vs. pred

Predictor: kde_feve



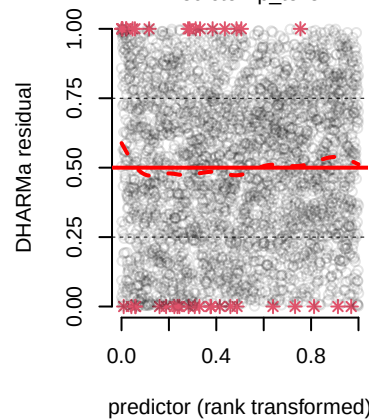
DHARMa residual Residual vs. pred

Predictor: tric



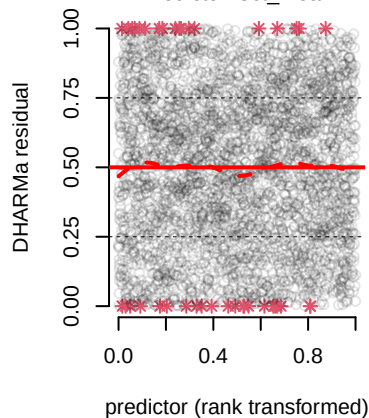
DHARMa residual Residual vs. pred

Predictor: p_teve



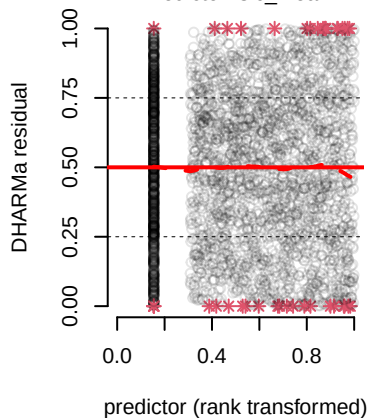
DHARMa residual Residual vs. pred

Predictor: sbt_mean



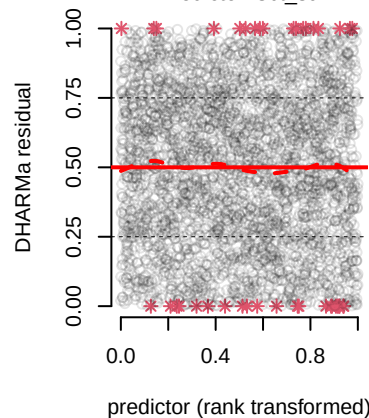
DHARMa residual Residual vs. pred

Predictor: sic_mean



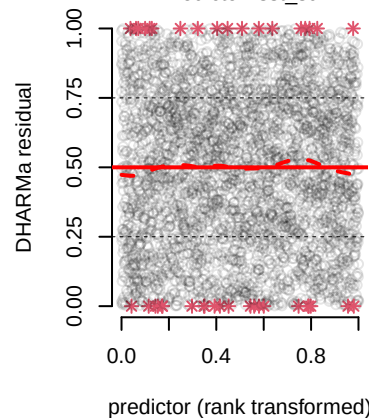
DHARMa residual Residual vs. pred

Predictor: sbt_sd



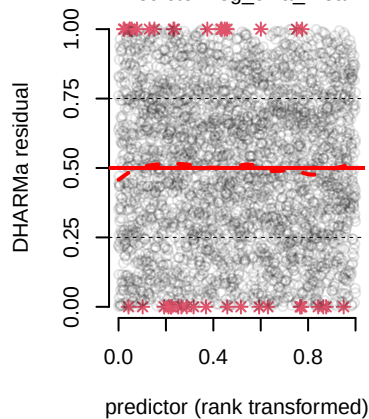
DHARMa residual Residual vs. pred

Predictor: sst_sd



DHARMa residual Residual vs. pred

Predictor: log_chla_mean



DHARMa residual Residual vs. pred

Predictor: depth

